

PRESIDENCY UNIVERSITY

School of Artificial Intelligence and Advanced
Computing



CSS7102- Mini Project Review-03

CIVIC-VOICE An AI .Assistant for Multi-Channel Citizen Feedback Analysis in Public Services

Batch Number: CAI-20

Under the Supervision of,

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20231CAI0117

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Problem Statement Number: 40

Organization: Public Service Organizations (Government Agencies)

Category : Software

Problem Description:

Citizen feedback is received through multiple channels such as government grievance portals, local complaint systems, social media, emails and surveys. These sources are heterogeneous and are often analyzed separately, making it difficult to identify common issues and emerging concerns.

Civic Voice aims to unify multi-channel citizen feedback and apply NLP-based analysis to identify sentiment, recurring topics, important complaints and emerging issues through an interactive dashboard.

Difficulty Level: Medium–High

Abstract:

CIVIC-VOICE is an AI-based system designed to analyze citizen feedback collected from multiple channels including government grievance portals, local complaints, surveys, social media and emails.

During the current implementation phase, heterogeneous datasets were collected, normalized and integrated into a unified master dataset. A text preprocessing pipeline was developed to clean and standardize citizen feedback and prepare it for downstream NLP analysis.

The current master dataset contains 205,168 feedback records from five sources. Data validation and exploratory analysis were performed to examine text quality, missing information, urgency distribution and other data-quality issues.

The processed dataset will be used for subsequent sentiment analysis, topic detection, issue classification, prioritization and dashboard development.

Github Link

Github Link:

<https://github.com/CivicVoice-AI/AI-Assistant-for-Multi-Channel-Citizen-Feedback-Analysis>



OBJECTIVES

- **Multi-channel integration**

Collect and integrate citizen feedback from multiple sources into a unified dataset.

- **Sentiment analysis**

Clean, normalize and standardize heterogeneous citizen-feedback datasets.

- **NLP preparation**

To identify recurring service-related topics and emerging issues from large volumes of unstructured citizen feedback using topic-analysis techniques.

- **Issue prioritization**

Identify high-priority citizen concerns based on available urgency and analytical indicators..

- **Decision-support dashboard**

Develop an interactive dashboard to visualize citizen concerns, trends and analytical results.

Analysis of Problem Statement

Technology Stack Components:

Component	Technology	Purpose
Frontend	Streamlit / HTML, CSS, JavaScript	User interface and dashboard
Backend	Python, Flask (or FastAPI)	API development and application logic
Programming Language	Python	Core development
NLP & AI	Hugging Face Transformers, BERT, NLTK, spaCy	Text preprocessing, sentiment analysis, summarization
Machine Learning	Scikit-learn	Complaint classification and model evaluation

Analysis of Problem Statement (contd...)

Component	Technology	Purpose
Database	MySQL	Store citizen feedback and analysis results
Data Processing	Pandas, NumPy	Data cleaning and manipulation
Visualization	Plotly, Matplotlib	Graphs, charts, and analytics
Development Environment	Jupyter Notebook, VS Code	Model development and coding
Version Control	Git & GitHub	Source code management and collaboration
Deployment (Optional)	Render / Railway / Streamlit Cloud	Deploy the web application

NLP PREPROCESSING

- NLP Text Preprocessing

Raw Feedback



Text Cleaning



Normalization



Whitespace



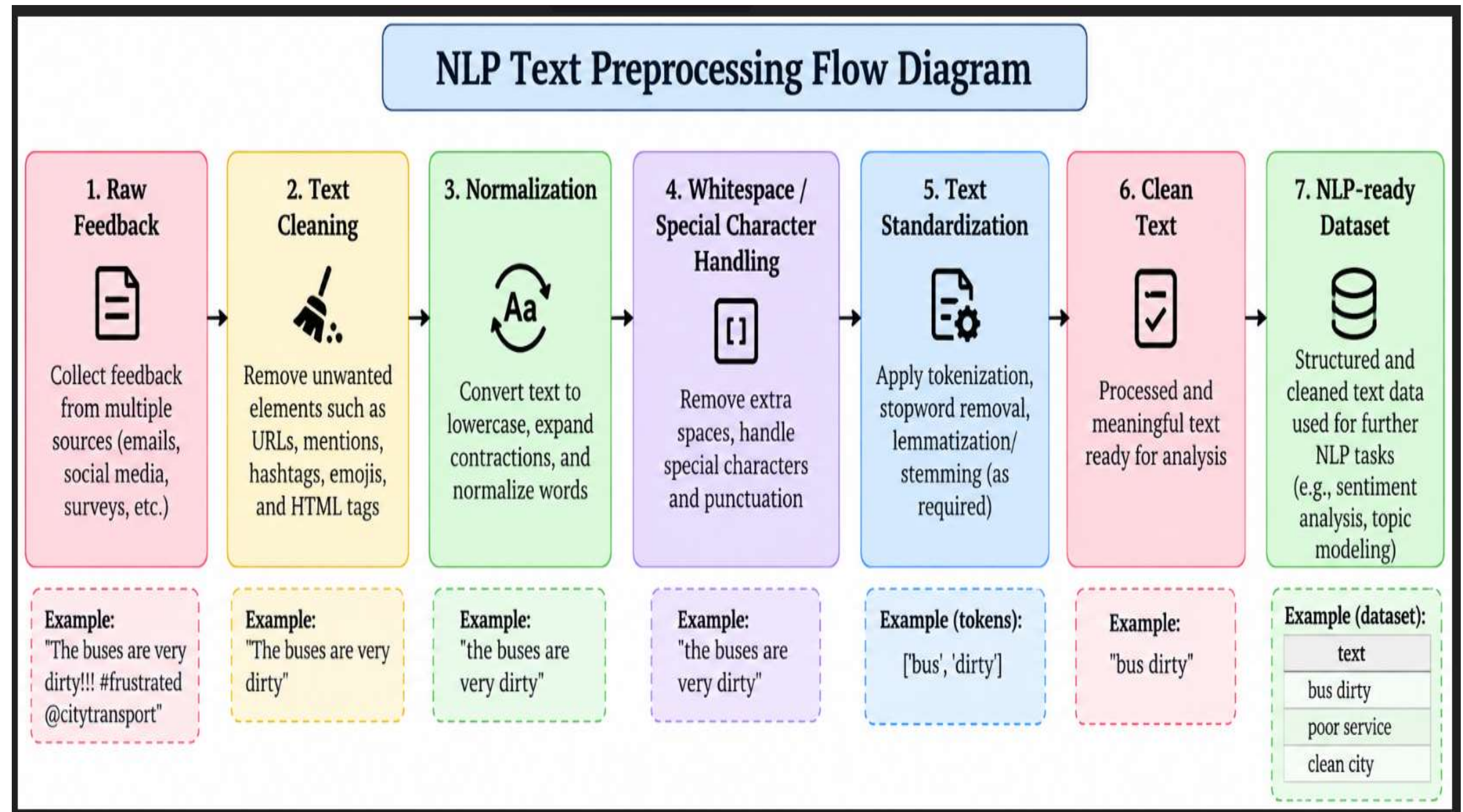
Text Standardization



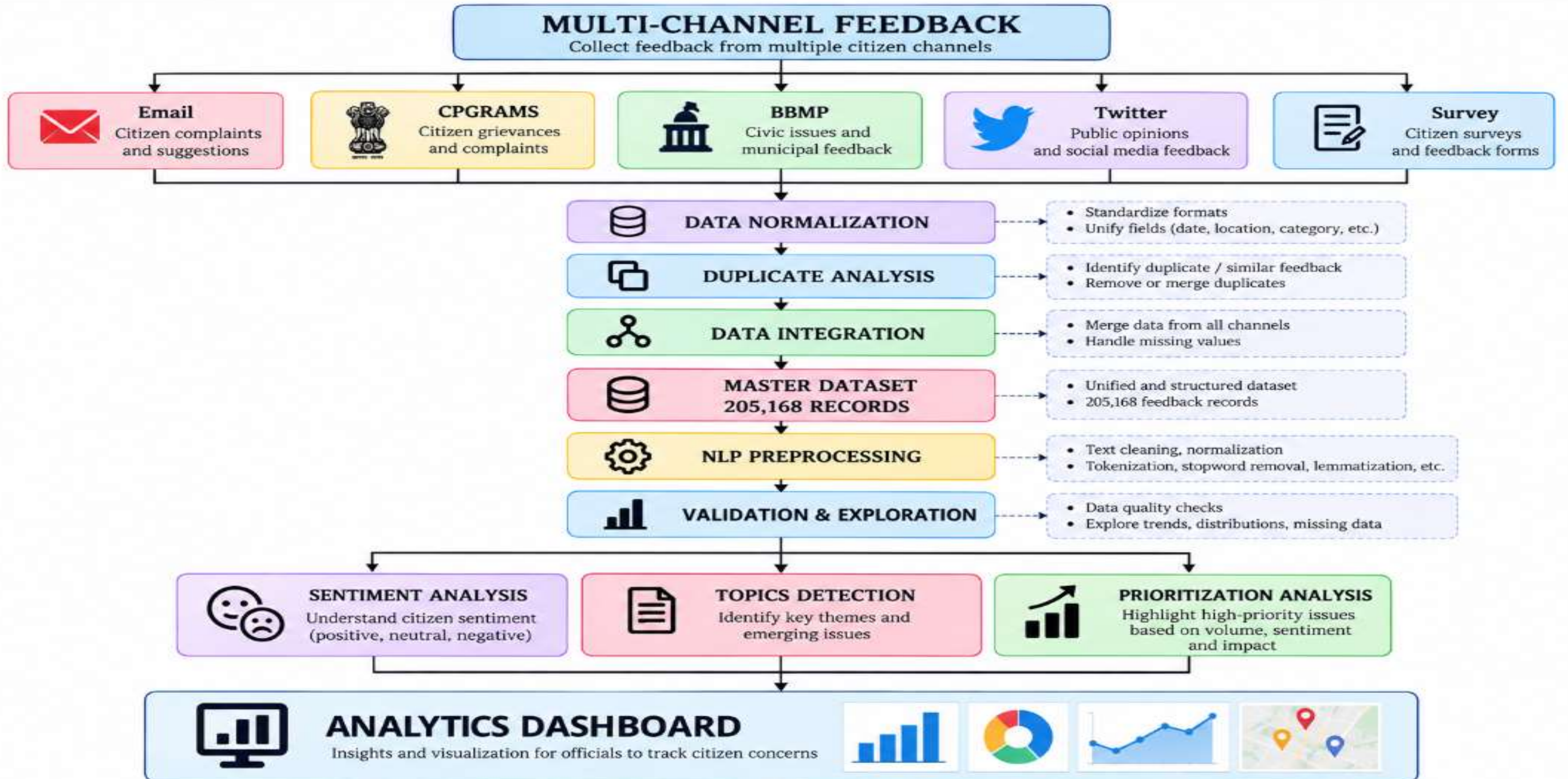
Clean Text



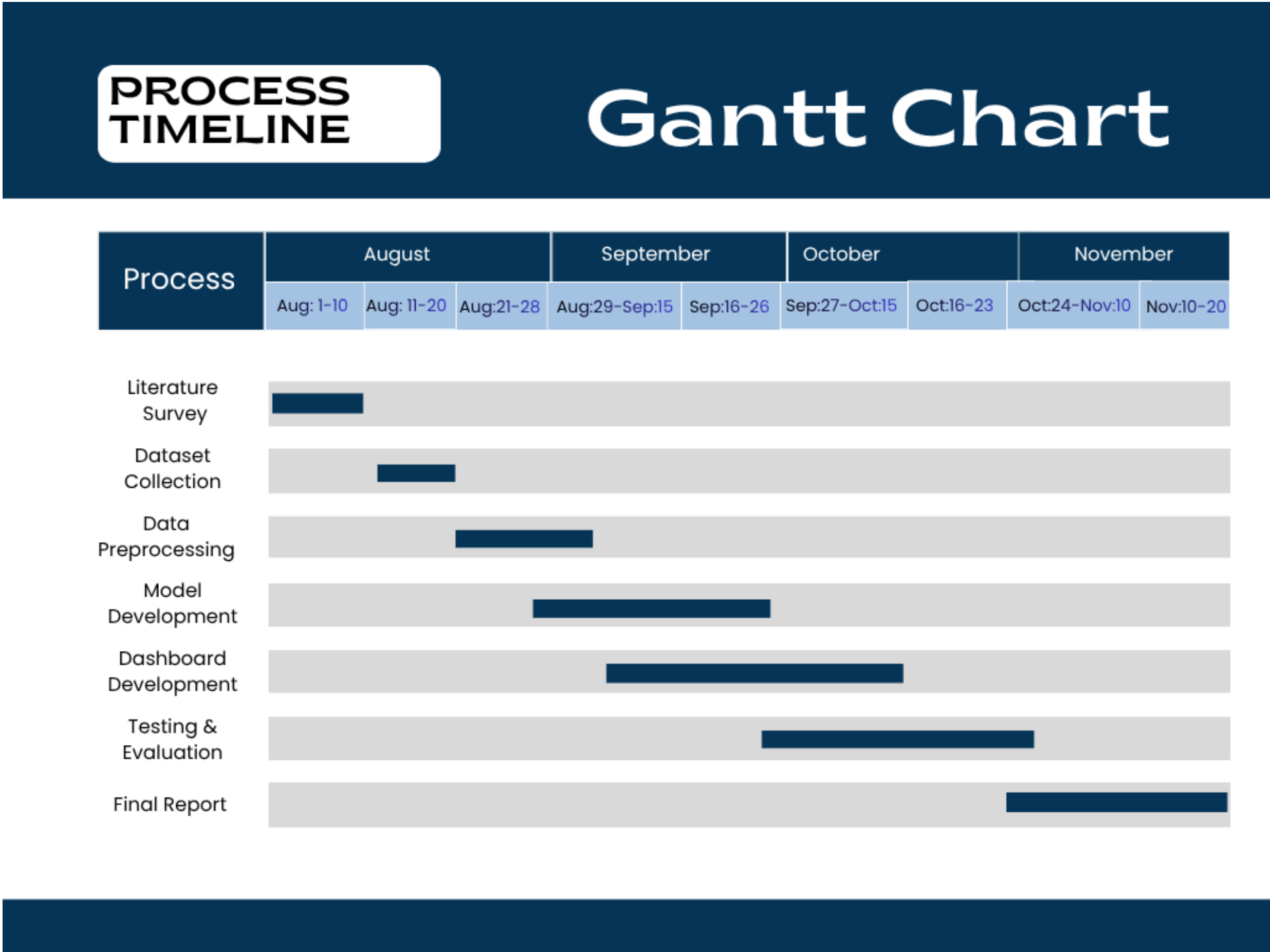
NLP-ready Dataset



Project Architecture



Timeline of the Project (Gantt Chart)



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Thank
You!

